

4.15 Hypersensitivity

▼ What is an adjuvant?

- A substance that **activates/enhances the body's immune response** against an antigen
- It allows the **body to provide a stronger immune response against a pathogen**

▼ What is a hapten?

- A small molecule which **on its own is not antigenic**, but when associated with a large carrier such as a **protein can form an antigen**.

▼ What is an immunogen?

- An antigen that can **independently elicit an immune response**

▼ What is an immune complex?

- An antibody bound to an antigen

▼ What is immunogenicity?

- The **ability of a substance to elicit an immune response**

▼ What is a damage associated molecular pattern (DAMP)?

- **Molecules that are released** as a result of non-apoptotic cell death
- They are recognised by and can **activate the innate immune system** (first line of defence) in order for it to fight against the causative agent of the damage

▼ What is a pathogen associated molecular pattern (PAMP)?

- **Common molecular patterns** found in **pathogens and microbes**, but **not host cells**
- These **patterns allow the innate immune system to recognise** that there are **pathogens** about

▼ What are PRRs (Pattern recognition receptors)?

- These are **proteins on hosts** that **recognise DAMPs and PAMPs**

▼ What is an antigen?

- Any molecule or molecular **structure** that **can be recognised by an antibody or the adaptive immune system**.

- ▼ Image showing the structures of the different types of antibody.

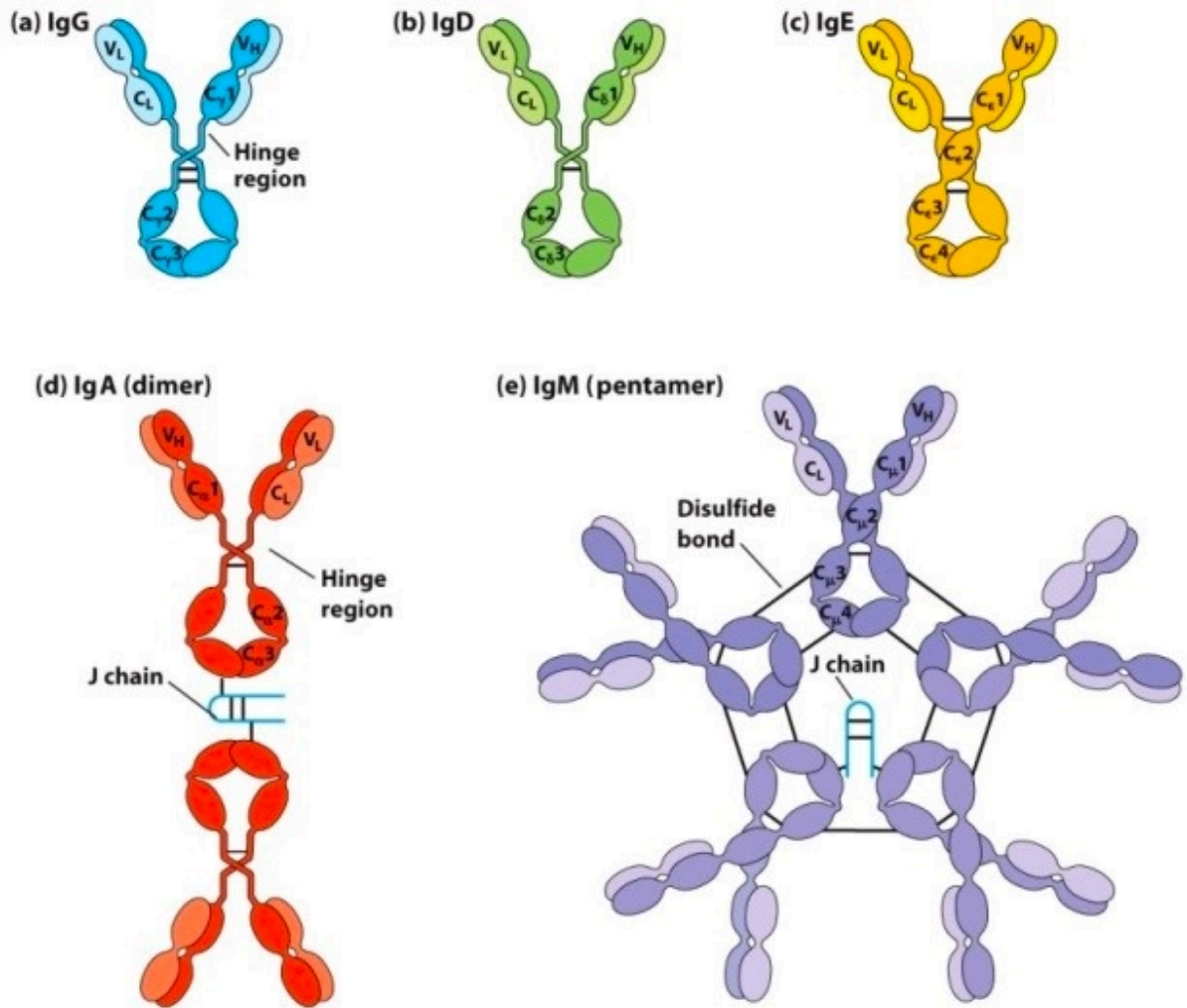


Figure 3-22
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- ▼ Which antibody potently activates complement and Fc receptor mediated phagocytosis?
 - IgG3
- ▼ Which antibody can cross the mucosal epithelium to deal with mucosal infections?
 - IgA
- ▼ Which antibody can induce mast cell degranulation and hence prompt release of inflammatory mediators?
 - IgE.

▼ Image showing T cell differentiation.

- As seen in this diagram, T cells are activated by their interactions with antigens
- This allows them to differentiate via clonal expansion into CD4 (helper, used to activate B cells and macrophages) or CD8 T cells (killer cells)

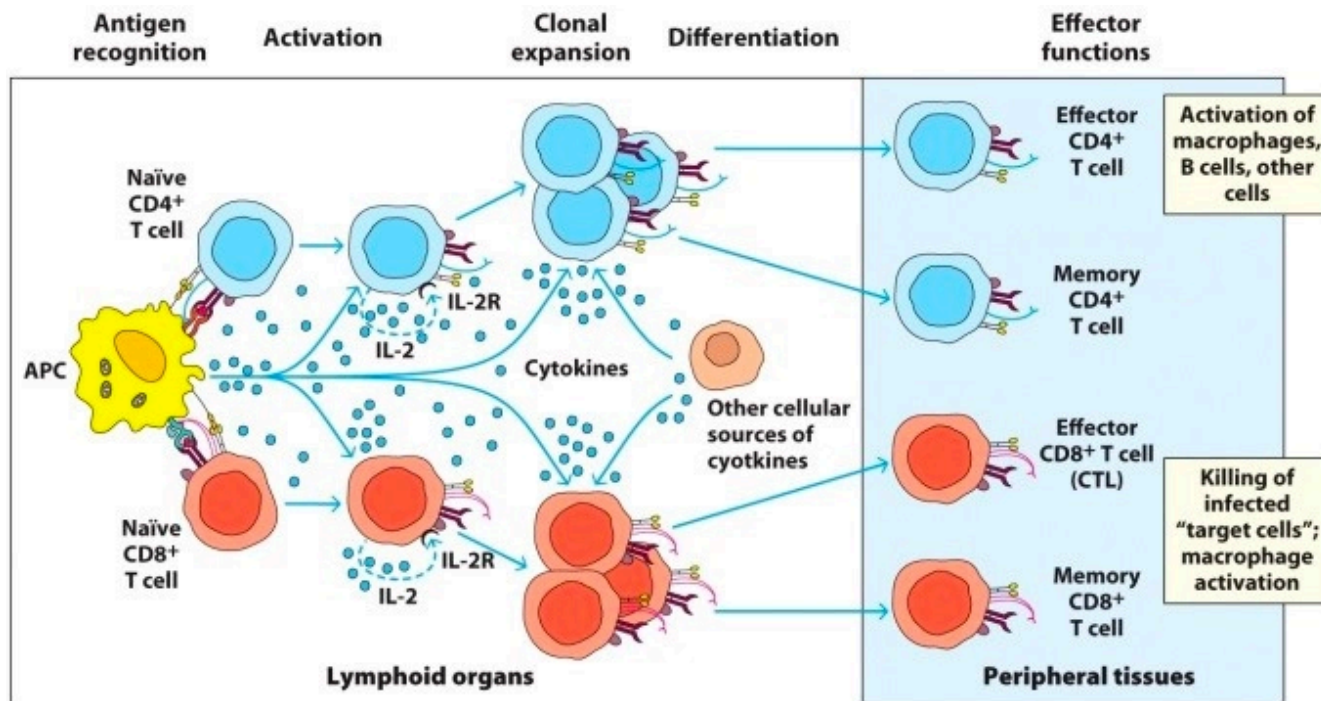


Figure 11-1

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- ▼ Image showing the different T helper cells that a naïve CD4 cell can differentiate into and their respective functions.

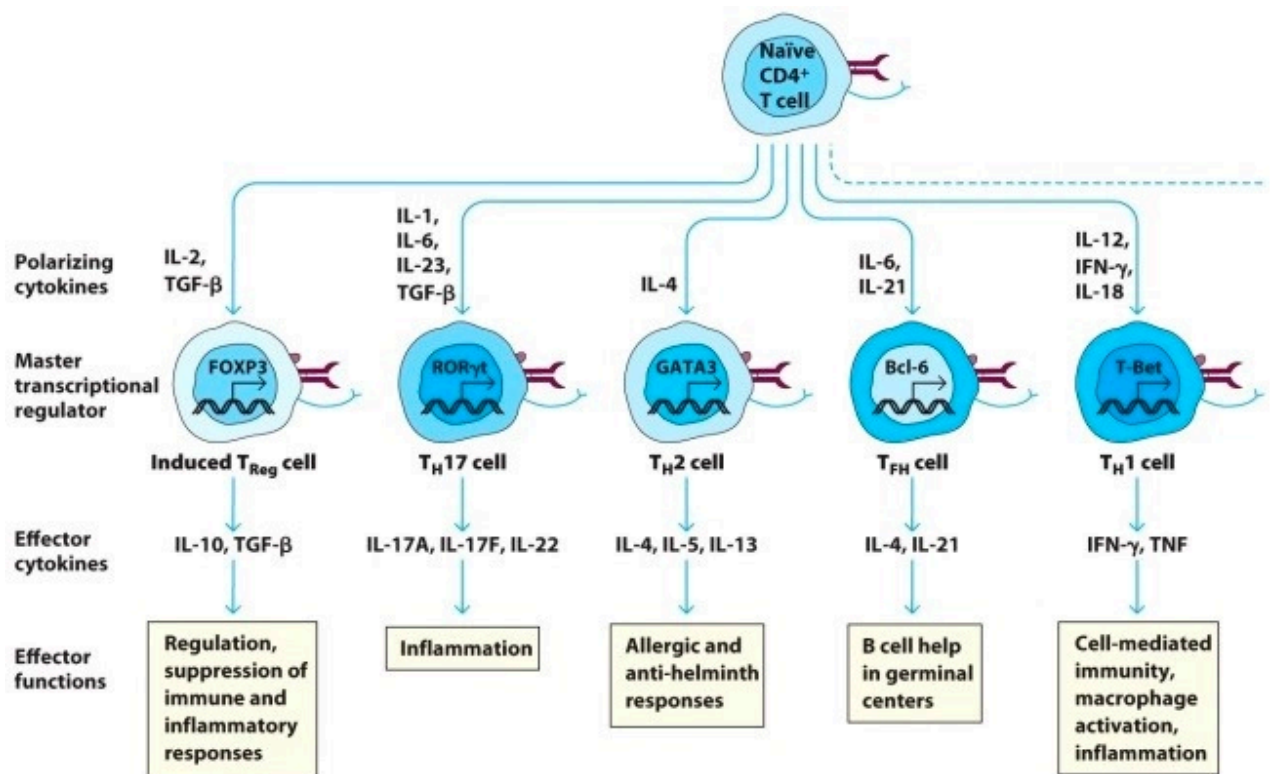


Figure 11-11
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- ▼ Which type of T cell is involved in the activation of macrophages, B cells and other cells?
- CD4 T cells
- ▼ Which type of T cell is involved in the killing of infected target cells?
- CD8 T cells
- ▼ Recall three diseases that result from Type I (immediate or anaphylactic) hypersensitivity reactions.
- Asthma
 - Allergic rhinitis (hay fever)
 - Atopic dermatitis (eczema)
- ▼ What can a skin prick test be used for in the context of allergy?
- Skin prick tests involve **exposing the skin to small amounts of allergen**
 - This can often be used to **diagnose allergy** as inflammation will occur in the case of an allergy

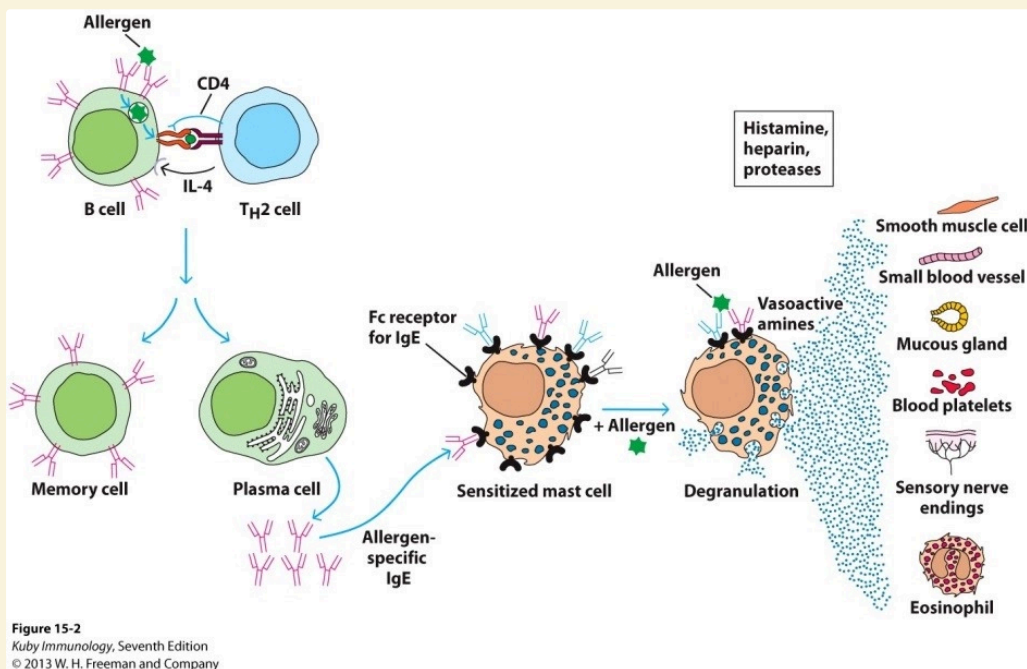
▼ What is a Type I hypersensitivity reaction?

- This is an **allergic reaction** that occurs when a **susceptible host is re-exposed to** what is known as an **allergen**
 - i.e they have already been exposed to the allergen before, and have antibodies in their arsenal that can work against this allergen
- It happens because **allergic individuals produce IgE antibodies against common antigens**, which shouldn't have antibodies produced against them in a normal situation
 - **IgE antibodies** are normally only **produced against potent antigens such as parasitic ones**
- It can lead to a **"wheal-and-flare" reaction** (i.e inflamed tissue)

▼ How is the immune system initially sensitised to a particular allergen?

- The initial sensitisation of the immune system to aberrantly recognise a harmless antigen as harmful is complex with factors in **genetics, age and the environment involved**

- ▼ Describe how IgE is initially produced in Type I hypersensitivity reactions.
 - The immune system is **sensitised to a particular allergen**, meaning it aberrantly recognises it as dangerous. This results in the production of:
 - **Th2 CD4 cells** and **T follicular helper (Tfh) CD4 cells**
 - These T cells **produce cytokines IL-4 and IL-13** and the action of these **cytokines on B cells** causes them to switch to producing **IgE specific to the allergen**
 - The **IgE then binds to mast cells** using its **Fc receptor**, leading to **degranulation of the mast cells**



- ▼ Which cells do IgE bind to and how do they do this?
 - **Basophils and mast cells**
 - These **granulocytic cells** express a **high affinity IgE receptor**, **Fc epsilon receptor 1 (FcεRI)**
- ▼ What are the two types of T helper cell produced in a Type I hypersensitivity reaction?
 - **Th2 CD4 T cells**
 - **T follicular helper (Tfh) CD4 T cells**
- ▼ What two molecules are produced by these cells?
 - **IL-4**
 - **IL-13**

- ▼ What effect do these molecules have in hypersensitivity reactions?
 - **IL-4 and IL-13** are cytokines that are used to **promote the production of IgE** when they act on B cells.
- ▼ How does IgE interact with antigens in a Type I hypersensitivity reaction, and what effect does this have on the cells around it?
 - **IgE then binds onto mast cells and basophils** due to their **high affinity for the Fc epsilon receptor I (FcεRI) found on IgE**
 - When **IgE is bound to antigens**, these **IgE are cross-linked and degranulated** and this leads to **release of inflammatory mediators such as histamine and serotonin**
 - The **speed and magnitude** of this response is **heightened during the inflammatory response** compared to normal infection
- ▼ Why are there three phases in Type I hypersensitivity reactions?
 - There is such a **wide range and variety of different molecules released** that the **release of these molecules can be classed into three phases**
- ▼ What happens in the early phase of a Type I hypersensitivity reaction?
 - **Bioactive small molecules** (i.e histamine, cytokines, leukotrienes and prostaglandins) are **produced directly by mast cells**
- ▼ What happens in the later phase of a Type I hypersensitivity reaction?
 - **Neutrophils are recruited** to the site of reaction
- ▼ What happens in the late phase of a Type I hypersensitivity reaction?
 - It happens **3-4 days after exposure** and it's where high frequencies of **eosinophils are recruited and Th2 cells are present.**
- ▼ What is the role of leukotrienes and prostaglandins in a Type I hypersensitivity reaction?
 - They are used to **cause smooth muscle to contract**
 - They are highly active molecules.
- ▼ Which two types of antibodies are involved in Type 2 hypersensitivity reactions?
 - **IgG and IgM**
- ▼ What is an alternative name for Type 2 hypersensitivity reactions?
 - **Antibody-mediated cytotoxic hypersensitivity**

▼ What happens in Type 2 hypersensitivity reactions?

- The **IgM or IgG antibodies** bound to self cell antigens, and sometimes go on to kill these cells
- These reactions are otherwise known as **antibody-mediated cytotoxic hypersensitivity** reactions

▼ What are the two ways that Type 2 hypersensitivity reactions come about?

- Exposure to a foreign antigen which is recognised by antibodies
- Aberrant (unusual) response of the antibodies against self-antigens

▼ What are the three ways that these reactions can cause disease?

- Anti-receptor activity – blocking or activating a receptor's function
- Antibody dependent cell-mediated cytotoxicity (abbreviated to ADCC)
- Classical activation of the complement cascade
 - This leads to inflammation, opsonisation and recruitment of immune cells

▼ Recall two examples of Type 2 hypersensitivity reactions.

- Immune thrombocytopenia (antibodies act against self platelet antigens)
- Graves disease
- Blood transfusion reactions

▼ What happens in Grave's disease?

- Patients develop **thyroid stimulating antibodies** that bind the **thyrotropin receptor** resulting in **secretion of thyroid hormones**.

▼ Where are antibody-antigen complexes bound to in Type 2 hypersensitivity reactions?

- Fc receptors (they bind to constant region of antibody) on the surface of Natural Killer (NK) cells and granulocytes
- This leads to the **direct lysis of the target cell**, but also release of inflammatory mediators, chemokines and cytokines
- Hence, Type 2 hypersensitivity reactions can also lead to **loss of function or imbalance in organ function**

- ▼ What is antibody-dependent cell-mediated cytotoxicity (ADCC)?
 - In **ADCC**, **antibody-antigen complexes on the surface of cells** are **bound by Fc receptors** expressed by cells such as **granulocytes and NK cells**
 - This leads to **directed lysis of the target cell**, but also the **release of inflammatory mediators, chemokines and cytokines**.
- ▼ What is an alternative name for Type 3 hypersensitivity reactions?
 - Immune complex driven disease
- ▼ What happens in Type 3 hypersensitivity reactions?
 - Immune complexes (**non cell-bound** antibody-antigen complexes) form and normally they are cleared up by the immune response
 - In **Type III HRs**, the **immune complexes aren't cleared up** and this leads to their **deposition in blood vessel walls and tissues**, promoting inflammation and tissue damage due to excess **neutrophil extravasation to the tissues**
 - i.e if **antibodies are produced against DNA**, since almost all cells have DNA, it will be near **impossible to clear up these immune complexes**
 - This can lead to **fever, rashes, joint pain or protein in the urine: vasculitis if deposited in blood vessels, glomerulonephritis if in the kidneys or arthritis if in the joints**.
- ▼ Describe the distinction between Type 2 and Type 3 hypersensitivity reactions.
 - **Type II** reactions involve **recognition of antigens on the surface of cells** by **IgGs and IgMs** and the **lysis or dysfunction of those cells**
 - **Type III** reactions, again using antibodies, recognise **antigen that is freely available in the body, not attached to cells** and usually present over sustained periods.
- ▼ Recall three autoimmune diseases caused by Type 3 hypersensitivity reactions.
 - Rheumatoid arthritis
 - Multiple sclerosis
 - Systemic lupus erythematosus (SLE)
- ▼ What happens in systemic lupus erythematosus (SLE) reactions?
 - IgG binds to DNA or nucleoproteins which form deposits

- ▼ How is anti-serum used to treat patients with snake bites and how can Type 3 hypersensitivity interfere with this?
 - Antibodies for the snake venom particles are injected into the body